

1 Benchmarking sequence and structure homology based 2 methods for eukaryotic virus protein annotation

3 Lander De Coninck^{1*}, Uri Neri² and Simon Roux²

4 ¹KU Leuven, Microbiology, Immunology and Transplantation, Clinical and
5 Epidemiological Virology, Herestraat 49, box 1040, Leuven, 3000, Belgium.

6 ²Lawrence Berkeley National Laboratory, DOE Joint Genome Institute, Cyclotron
7 Road 1, Berkeley, CA 94720, USA.

8 *Corresponding author(s). E-mail(s): lander.deconinck@kuleuven.be;

9 Contributing authors: uneri@lbl.gov; sroux@lbl.gov;

10 Abstract

11 Introduction

12 Results & Discussion

13 Dataset overview

- 14 • general overview of data and structures
 - 15 – TODO: kingdom distribution (proteins vs species)
- 16 • performance of methods in hits
- 17 • comparison of methods in shared hits
- 18 • comparison of functional annotation per method to original
- 19 • how many are informative annotations
 - 20 – TODO: which conversions from hypothetical protein to informative (phages?)
- 21 • computational benchmark (roughly)

22 Figure 1

23 Figure 2

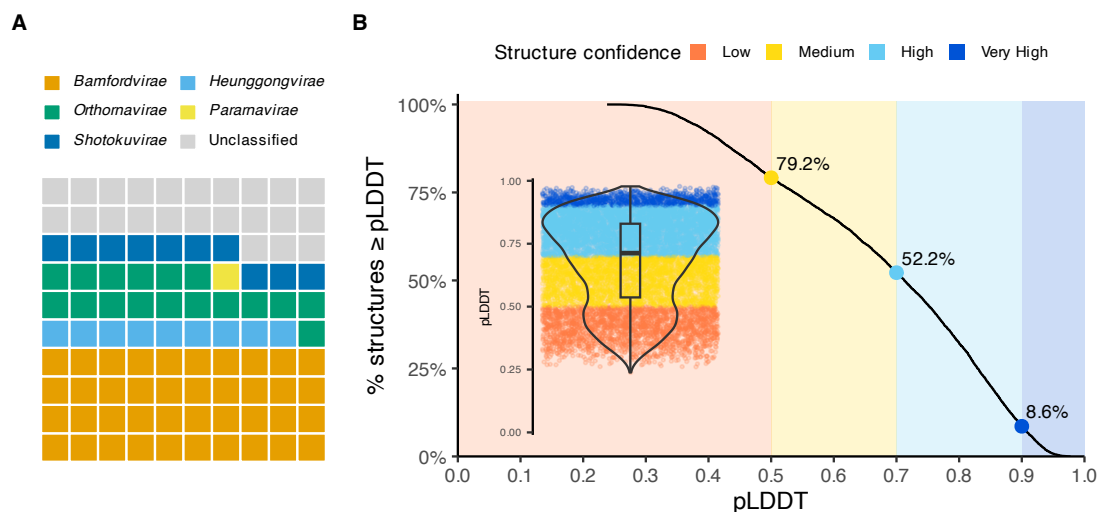


Figure 1: A) Waffle plot showing the distribution of the number of proteins from the different viral Kingdoms. Each tile roughly corresponds to 110 proteins. B) A cumulative distribution function plot showing the percentage of predicted structures from the test set with a pLDDT score above the pLDDT score on the x-axis. The inset shows a violin- and boxplot with all pLDDT scores from the predicted structures (n=11,467).

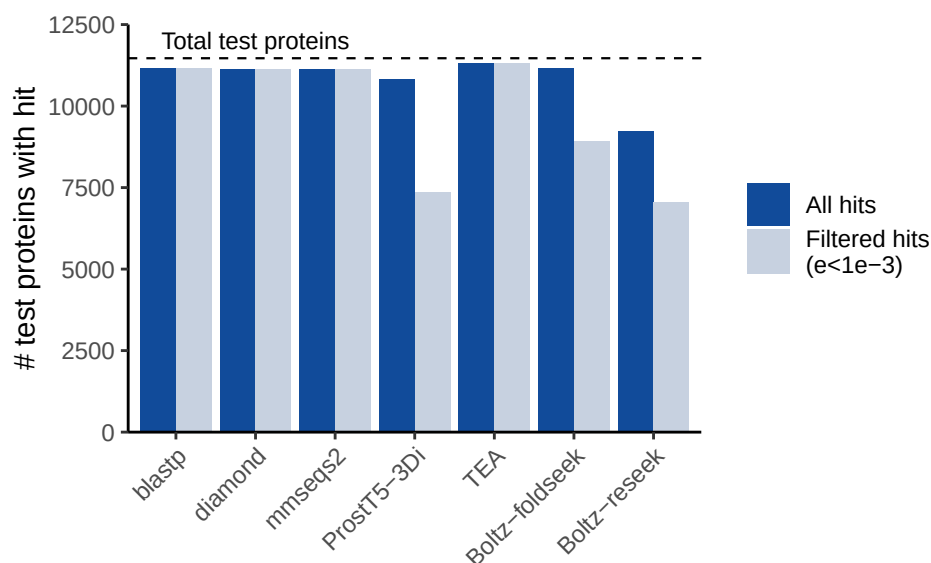


Figure 2: Bar plot showing the number of proteins that got a hit against the BFVD shown per method before and after filtering the reported results on e-value lower than $1e-3$.

Conclusion

Material and Methods

Dataset selection

Test set

To benchmark different methods for eukaryotic virus protein annotation, we selected 10% of the exemplar species in ICTV's Virus Metadata Resource (VMR) v40.2 that did not have bacteria

30 or archaea as designated host (n=1,000). We subsequently extracted their GenBank accession
31 numbers (n=1,402) and downloaded all associated proteins (n=11,360).

32 Protein sequences that were longer than 2,500 amino acids were split into multiple chunks of
33 maximum 2,500 amino acids so that structure prediction with Boltz-2¹ would not run out of
34 GPU memory (see below). This resulted in a final test set of 11,467 protein sequences from
35 1,000 eukaryotic virus species.

36 Database

37 The BFVD² was used as the reference database for both sequence and structure homology-
38 based annotation. The BFVD is a comprehensive database of predicted protein structures from
39 the viral sequence representatives of the UniRef30 clusters. UniRef30 is a UniProt database that
40 clusters protein sequences at 30% sequence identity and selects a representative sequence
41 for each cluster, these representatives thus have a UniProt accession. Since a substantial
42 amount of BFVD's included proteins are no longer available in UniProtKB as they are not part
43 of reference proteomes, we converted their UniProt accessions to UniParc IDs.

44 Functional categories for each BFVD entry were derived from protein names associated with
45 the corresponding UniParc IDs (could be from UniProtKB/TrEMBL, RefSeq and/or GenBank)
46 for eukaryotic viruses (also see below) or by classifying them into PHROG-like³ categories with
47 Empathi⁴ in the case of bacteriophage proteins. The distinction between eukaryotic virus and
48 bacteriophage proteins was made based on the taxid information from the UniParc protein ID.

49 The BFVD database was further converted into the required formats the search tools used in
50 this benchmarking effort (see sequence and structure homology-based annotation).

51 Functional classifier

52 Functional categories for eukaryotic virus proteins were designed with Perplexity AI in deep
53 research mode using the following prompt: "Can you give me an exhaustive list of common
54 protein functions in eukaryotic viruses? Use <https://viralzone.expasy.org> and <https://ictv.global/report/genome> as sources. Give me a machine readable file with all these functional
55 categories." The resulting functional categories were further manually curated to ensure that
56 they were mutually exclusive. The final list of functional categories is available in **Supplementary**
57 **Table 1**.

58
59 A custom script generated by Perplexity AI was used to assign functional categories to BFVD
60 proteins based on all their protein names from UniParc in a JSON file. The script uses pattern
61 matching to assign the most appropriate functional category to each protein. If no keywords
62 match, the protein is assigned to the "Hypothetical Protein" category. The prompt to generate
63 the classifier script was: "Write me a script that classifies the proteins in the uploaded JSON file
64 (as specific as possible) into these functional categories, also if there are proteins in this file that

do not fit in one of your proposed categories search new sources and create new functional categories.”. Followed up by: “Split the polymerase category in RNA-dependent RNA polymerase, DNA polymerase and reverse transcriptase. In addition split the Capsid/Nucleocapsid/Core Structural into Capsid and Nucleocapsid categories. Split helicase and NTPase.”.

The pattern matching rules, functional categories, and classification script were subsequently refined through manual curation to optimize the functional assignment based on protein names.

The same functional classifier script was also used on the protein names from our test set to assign functional categories to the eukaryotic virus proteins downloaded from the ICTV exemplar species.

Sequence homology-based annotation

For sequence homology annotation, we used three state-of-the-art tools widely used in functional annotation pipelines and sequence similarity searches. As gold standard, blastp⁵ was used with an e-value cutoff of 1e-3. In addition, we used diamond⁶ in sensitive mode and mmseqs2⁷ with default settings; both have a default e-value cutoff of 1e-3.

Structure prediction

For the structure prediction of our test set proteins, we relied on Boltz-2¹ (default settings) with multiple sequence alignments (MSAs) generated by a local, adapted ColabFold installation⁸ (see <https://github.com/gbouras13/colabfoldv>). To increase the depth and diversity of the MSAs, we used the default ColabFold sequence databases, supplemented with extra *Orthornavirae* sequences⁹ and viral sequences used in the structure prediction for the Phold database¹⁰. Structures were predicted on an NVIDIA H100 GPU with 80GB of VRAM.

Structure homology-based annotation

ProstT5

TEA

Foldseek

Reseek

Computational benchmarking

- hyperfine

1. Passaro, S. et al. Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. 2025.06.14.659707 (2025) doi:[10.1101/2025.06.14.659707](https://doi.org/10.1101/2025.06.14.659707).

- 94 2. Kim, R. S., Levy Karin, E., Mirdita, M., Chikhi, R. & Steinegger, M. [BFVD—a large repository of predicted viral protein structures](#). *Nucleic Acids Research* **53**, D340–D347 (2025).
- 95 3. Terzian, P. *et al.* [PHROG: Families of prokaryotic virus proteins clustered using remote homology](#). *NAR Genomics and Bioinformatics* **3**, lqab067 (2021).
- 96 4. Boulay, A., Leprince, A., Enault, F., Rousseau, E. & Galiez, C. [Empathi: Embedding-based phage protein annotation tool by hierarchical assignment](#). *Nature Communications* **16**, 9114 (2025).
- 97 5. Altschul, S. F., Gish, W., Miller, W., Myers, E. W. & Lipman, D. J. [Basic local alignment search tool](#). **215**, 403–410 (1990).
- 98 6. Buchfink, B., Reuter, K. & Drost, H.-G. [Sensitive protein alignments at tree-of-life scale using DIAMOND](#). *Nature Methods* **18**, 366–368 (2021).
- 99 7. Steinegger, M. & Söding, J. [MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets](#). *Nature Biotechnology* **35**, 1026–1028 (2017).
- 100 8. Mirdita, M. *et al.* [ColabFold: Making protein folding accessible to all](#). *Nature Methods* **19**, 679–682 (2022).
- 101 9. Mutz, P. *et al.* [The protein structurome of Orthornavirae and its dark matter](#). *mBio* **16**, e03200–24 (2024).
- 102 10. Bouras, G. *et al.* [Protein structure-informed bacteriophage genome annotation with Phold](#). *Nucleic Acids Research* **54**, (2026).